Telling similar entities apart as knowledge changes

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Defense planned for March 2027

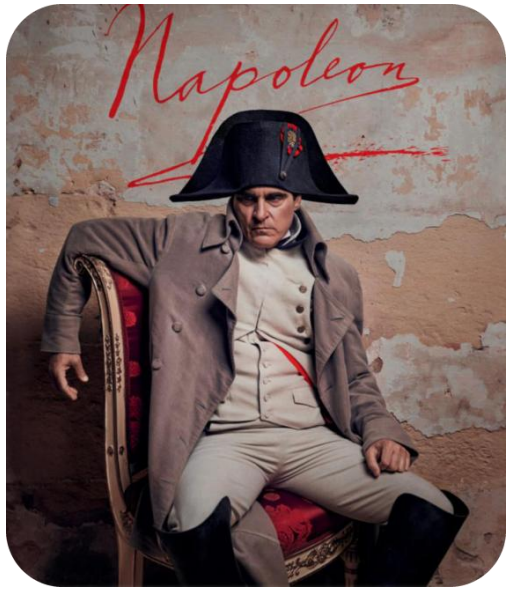
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Entity ambiguity and temporal drift



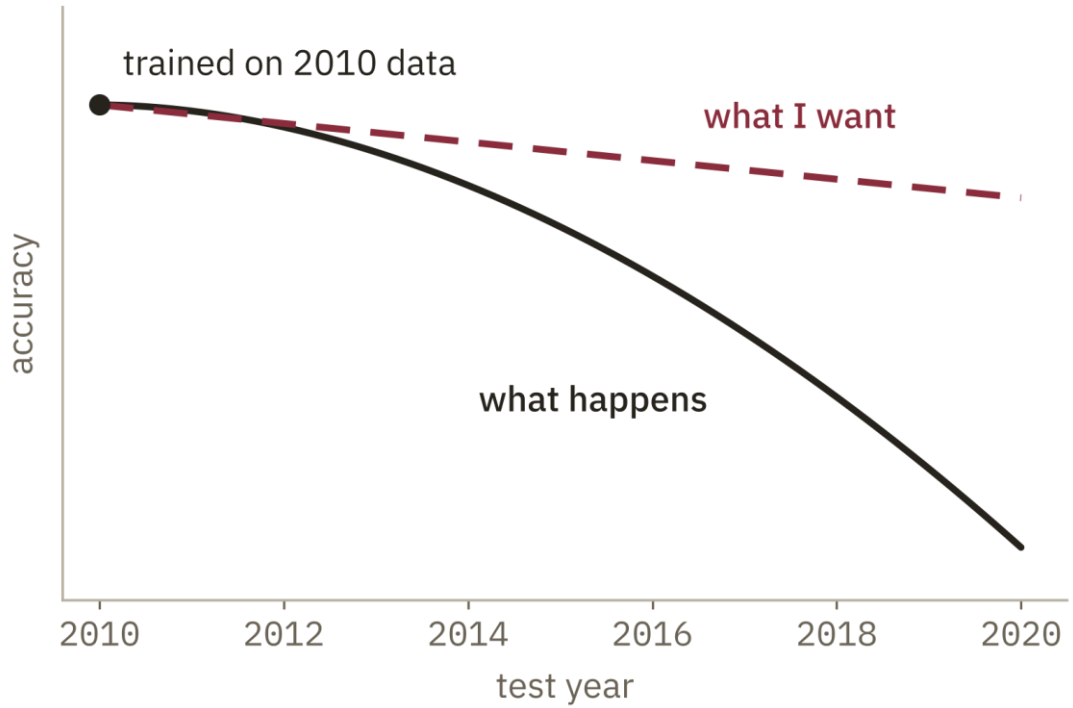
Napoleon Bonaparte

French Revolution
French Republic
Marie Antoinette

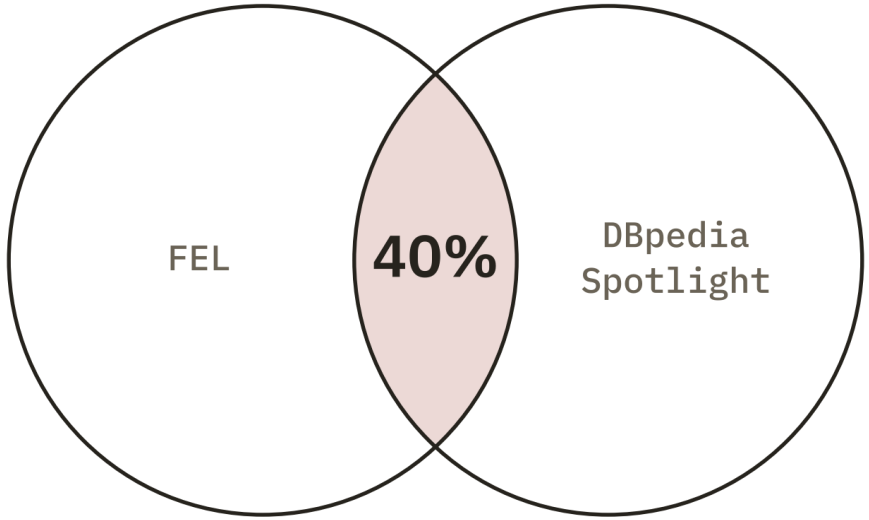


Napoleon (2023 film)

French Revolution
French Republic
Marie Antoinette

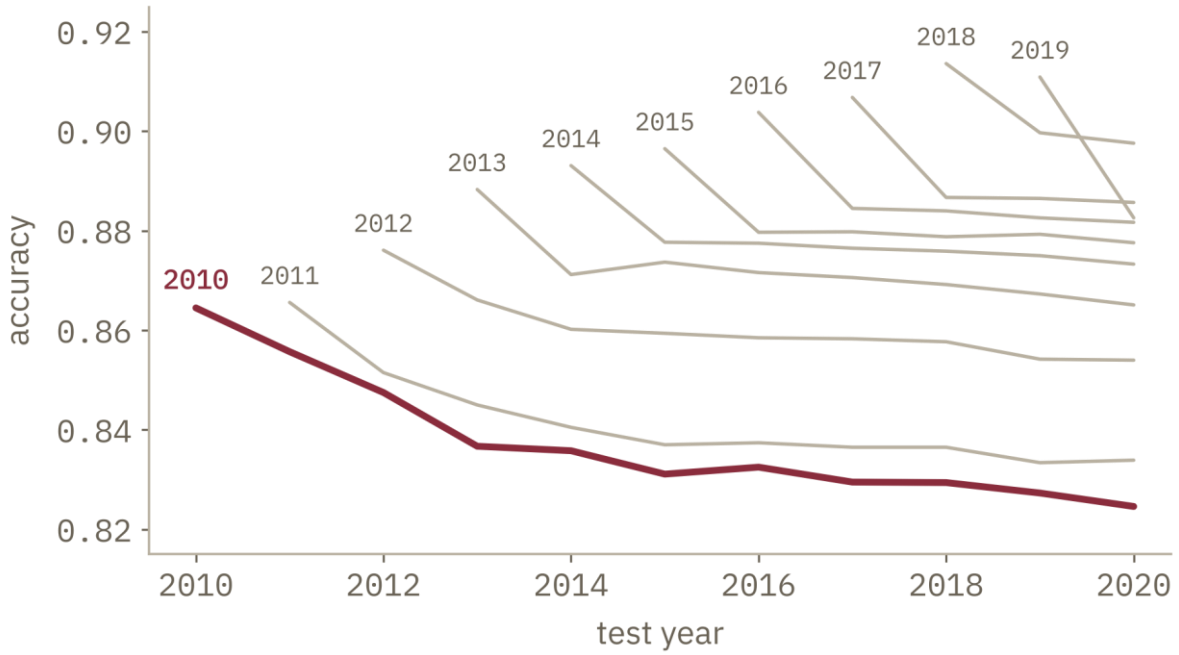


Real data shows both



found by both

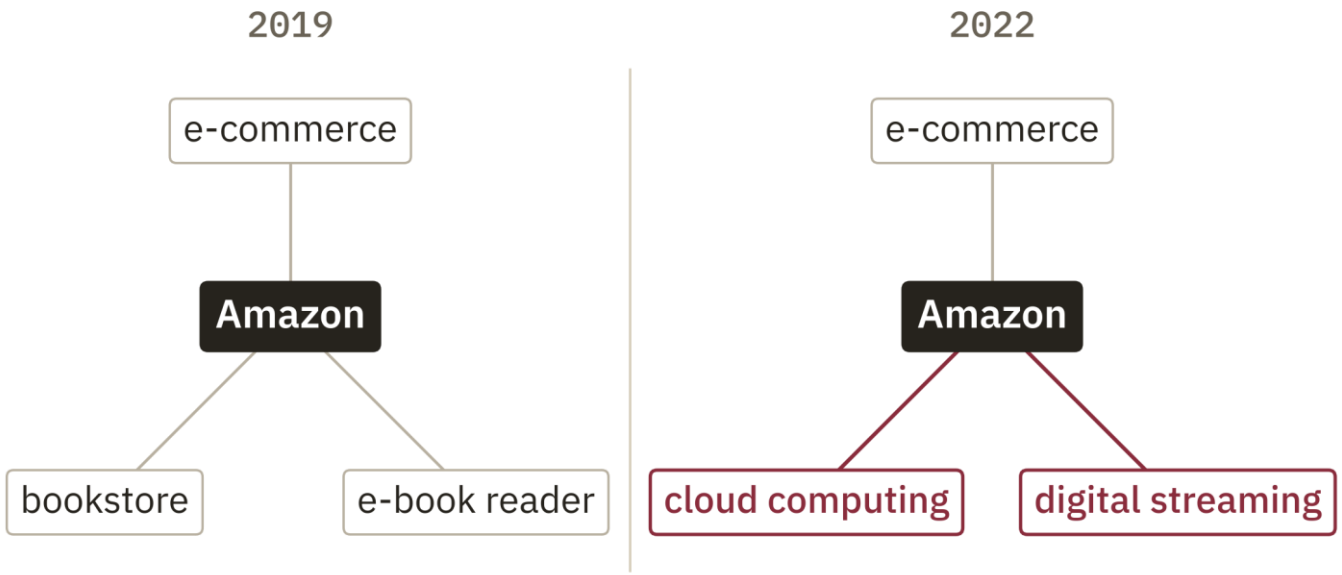
Understanding EL EKAW 2024



Graph-TempCZ LREC 2026

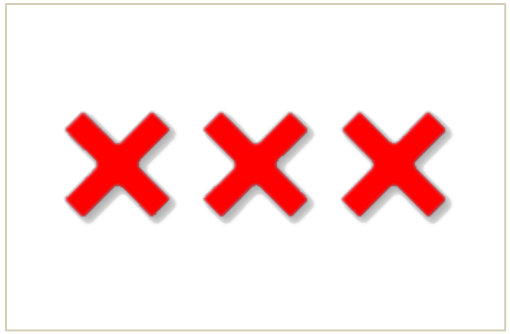
The graph and its changes

TIGER ECAI 2024 CYCLE CIKM 2024



TASK	Entity linking
DATA	Wikipedia text and the Wikidata graph, one snapshot a year, 2019-2022
RESULT	TIGER +21% and CYCLE +18% over the 2023 state of the art, three years after training

Semantics in discarded images



→ *Three red Saint Andrew's crosses arranged vertically, the emblem of Amsterdam.*



→ *An abstract cityscape of Amsterdam's canals with bold blue, red, and yellow blocks, stylized lampposts, and a bridge.*

TASK	Link prediction in multi-modal knowledge graphs
DATA	All Wikipedia images of each entity
RESULT	+5-8% Hits@1 across four models. On 20 logo entities, Hits@1 goes from 7.5 to 32.5

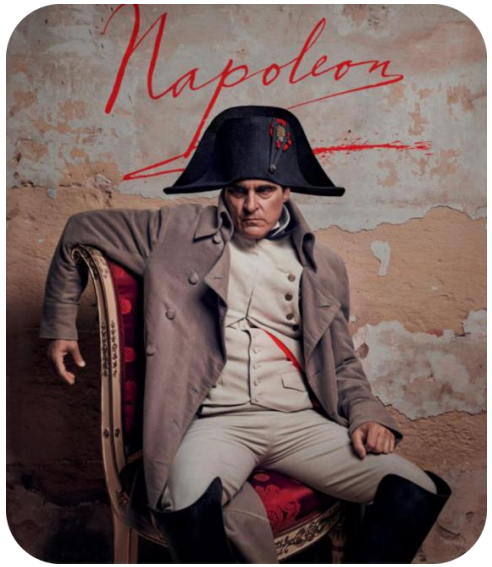
Time as a modality

Time Imprint Under review



Napoleon Bonaparte

1769, 1799, 1804, 1814, 1821



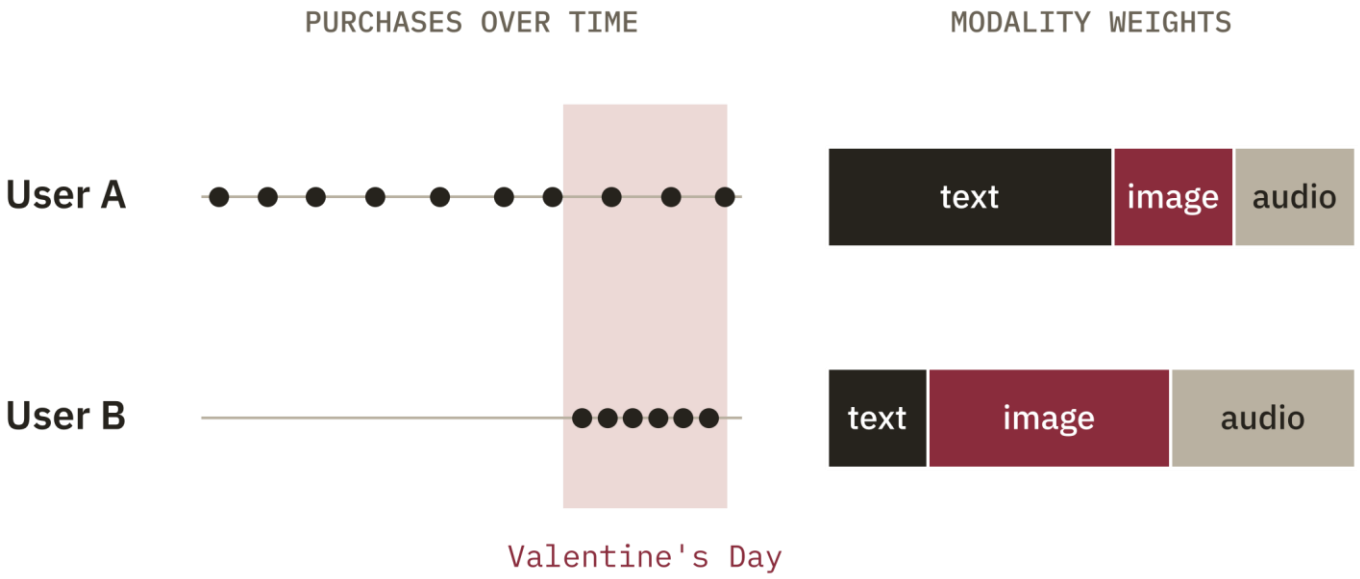
Napoleon (2023 film)

2020, 2023, 2024

TASK	Link prediction in multi-modal knowledge graphs
DATA	Years from each entity's description and image metadata
RESULT	+6% Hits@1 overall, +58% on the hardest 1% of look-alike entities

Each user’s temporal profile

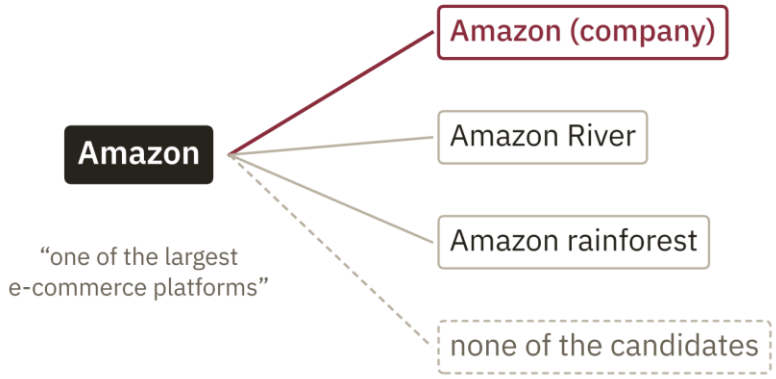
TimeRoute Under review



TASK	Multi-modal recommendation
DATA	Interaction histories from TikTok and Amazon
RESULT	+5-10% over the strongest baseline on three datasets, ten seeds

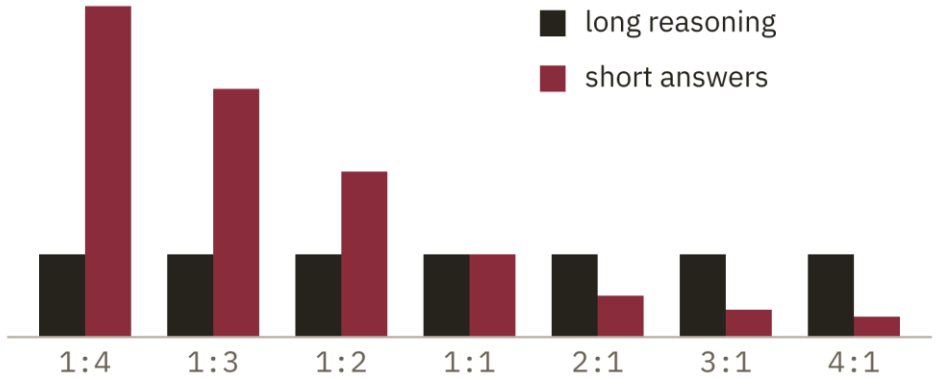
Two studies with LLMs

Select, Don't Train ISWC 2026



TASK	Entity disambiguation
DATA	News, tweets, Reddit and web text
RESULT	86.3 vs 82.3 F1 against the best trained dual encoder, with no retriever training

Fusion Training ACL 2026 SRW



TASK	Fine-tuning a hybrid reasoning LLM
DATA	Maths problems with short and long answers
RESULT	More short answers, weaker long reasoning (r = -0.58). Interleaving is the most robust schedule

Takeaways

1. Extra signal helps most where the model was weakest
2. The signal is often already in the data
3. An average hides both the problem and the gain

Publications

MASTER'S, BEIJING UNIV. OF TECHNOLOGY		PHD THESIS, UNIV. OF AMSTERDAM		PHD COLLABORATIONS	
DualChannel	Information 2021	TIGER	ECAI 2024	Understanding EL	EKAW 2024
Visual Analysis	JCAD 2022	CYCLE	CIKM 2024	Graph-TempCZ	LREC 2026
MVMA-GCN	EAAI 2023	Beyond Images	ESWC 2026	LLMs for graph data	ESWA 2026
RUNCL	Physica A 2024	Time Imprint	Under review	Fusion Training	ACL 2026 SRW
		TimeRoute	Under review	Select, Don't Train	ISWC 2026

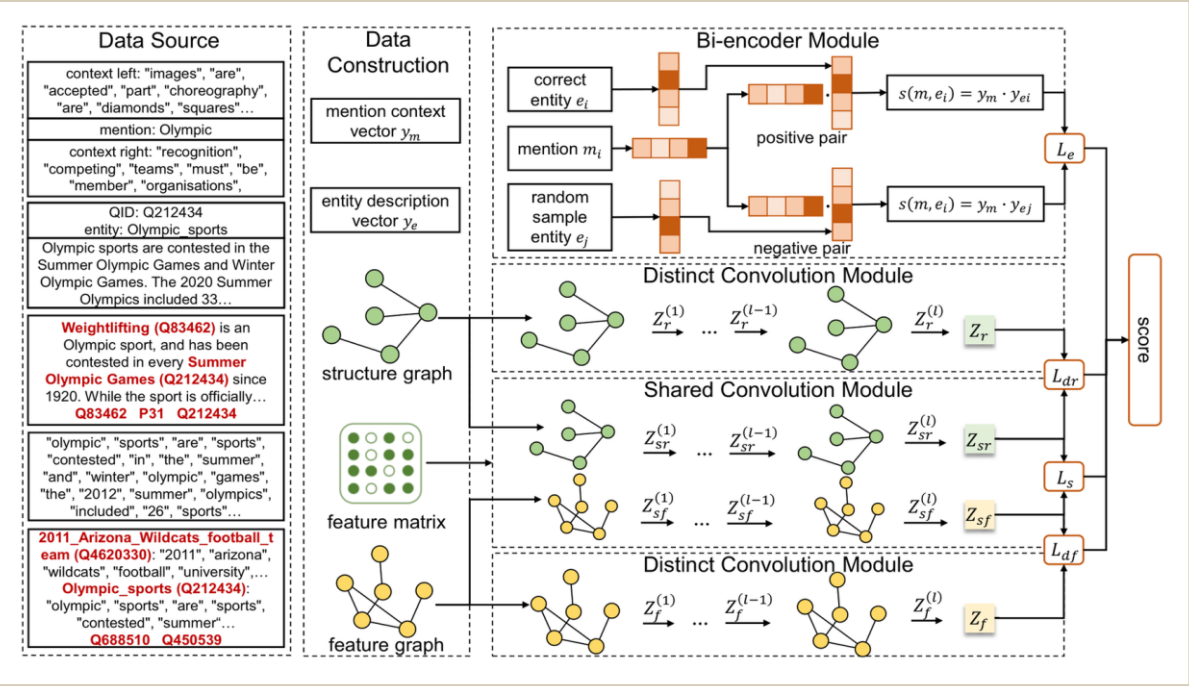
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TIGER

ECAI 2024

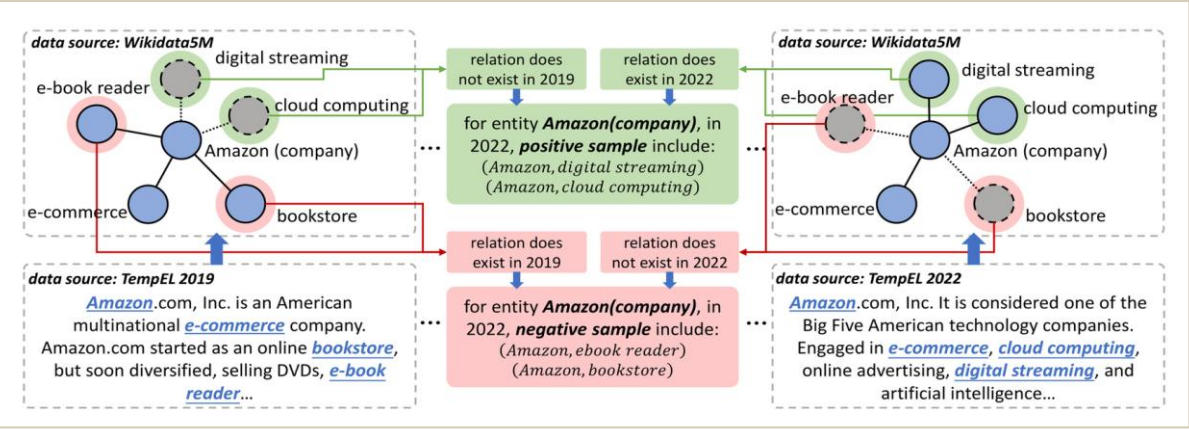


Pengyu Zhang, Congfeng Cao, Paul Groth. TIGER: Temporally Improved Graph Entity Linker. ECAI 2024.

PROBLEM	Text-only entity linkers get worse as the knowledge graph changes.
METHOD	I add two graphs during training, Wikidata links and description similarity. Shared and separate GCN layers with an HSIC loss keep what the graphs share and what is unique in each. After training the model reads text only.
RESULT	17% ahead of SpEL with no gap and 21% at a three-year gap, on entities present in every snapshot. Entities with more links gain more.
DATA	Graph-TempEL, which I built: 136k entities in four yearly snapshots, 2019-2022.

CYCLE

CIKM 2024

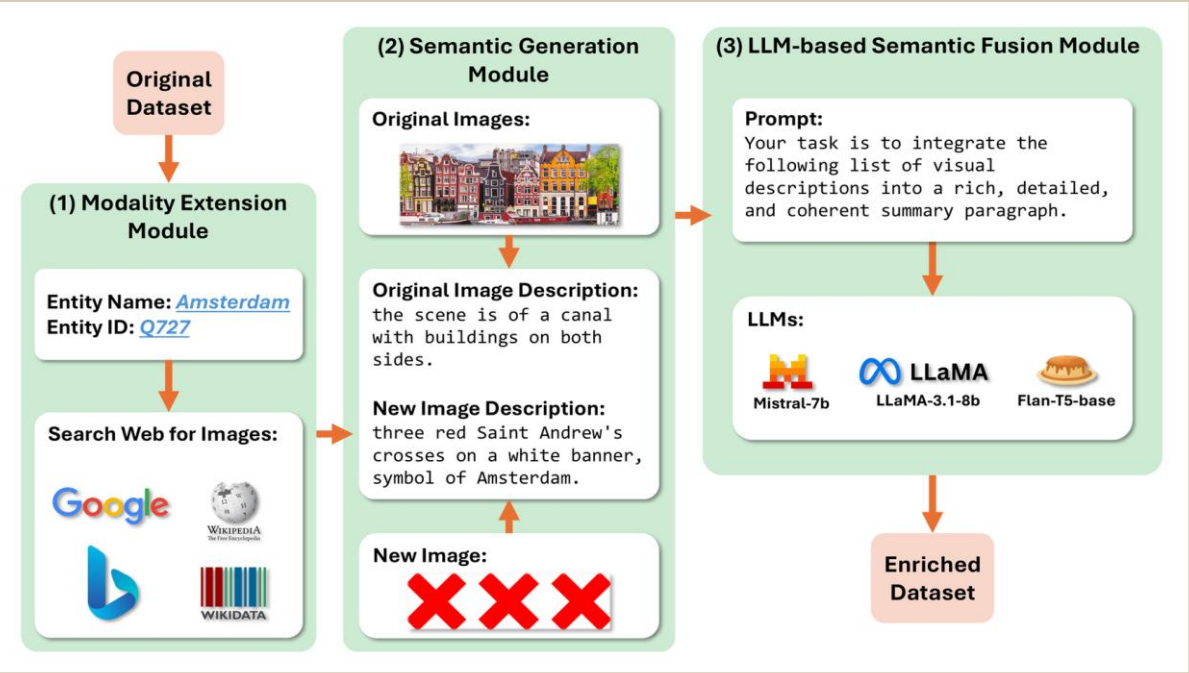


Pengyu Zhang, Congfeng Cao, Klim Zaporojets, Paul Groth. CYCLE: Cross-Year Contrastive Learning in Entity-Linking. CIKM 2024.

PROBLEM	TIGER treats each year's graph as fixed, so the changes between years go unused.
METHOD	Links that appear between two snapshots become positives, and links that disappear become negatives. A contrastive loss works across the relation graph and the feature graph.
RESULT	14% over SpEL after one year and 18% after three, on continual and new entities together. The largest gains are on low-degree entities.
DATA	GCL-TempEL: the same 136k entities, with the cross-year samples.

Beyond Images

ESWC 2026

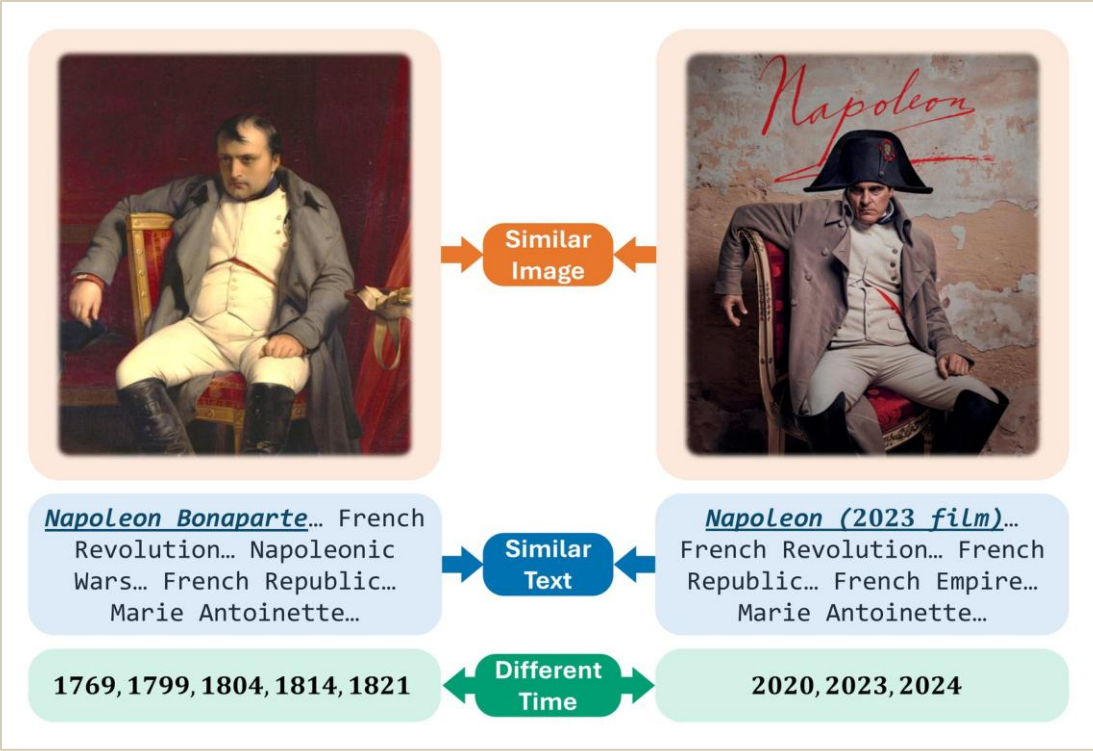


Pengyu Zhang, Klim Zaporojets, Jie Liu, Jia-Hong Huang, Paul Groth. Are a Thousand Words Better Than a Single Picture? Beyond Images - A Framework for Multi-Modal Knowledge Graph Dataset Enrichment. ESWC 2026.

PROBLEM	Curated multi-modal knowledge graphs drop ambiguous images such as logos, because their visual embeddings are close to noise.
METHOD	I collect all Wikipedia images of an entity, describe each one with a vision-language model, and fuse the descriptions with an LLM into one summary that is used as text.
RESULT	Four models on three datasets all improve, +5-8% Hits@1. On 20 logo and symbol entities Hits@1 goes from 7.5 to 32.5.
COST	The enrichment runs once, offline. The models and their losses stay as they are.

Time Imprint

Under review

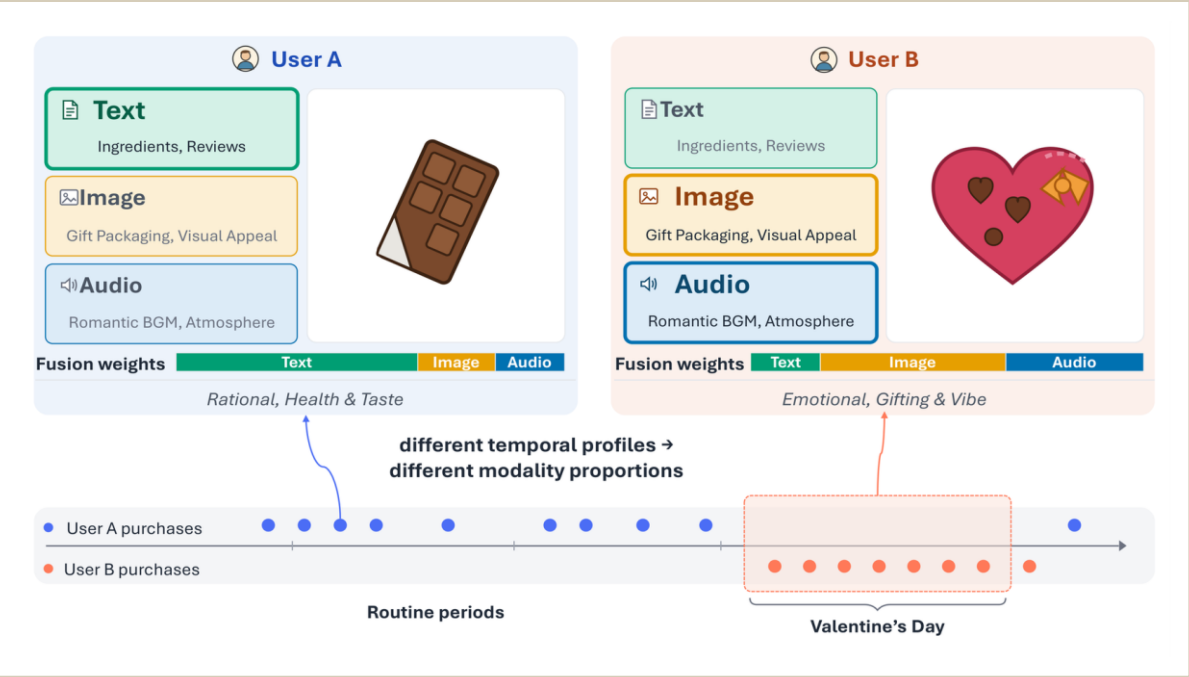


PROBLEM	Look-alike entities share text and images. Temporal knowledge graphs attach time to triples, so an entity has no time of its own.
METHOD	I take a few years around the median plus the earliest one, pool them with attention, and inject them into the encoder, the scoring function and a contrastive view.
RESULT	+6% Hits@1 on MKG-W, and +58% on the 1% most ambiguous entities. With 40% of the timestamps corrupted, Hits@1 on MKG-W drops by 0.7 points.

Pengyu Zhang, Klim Zaporojets, Congfeng Cao, Jia-Hong Huang, Paul Groth. Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs. arXiv 2607.09777, under review.

TimeRoute

Under review

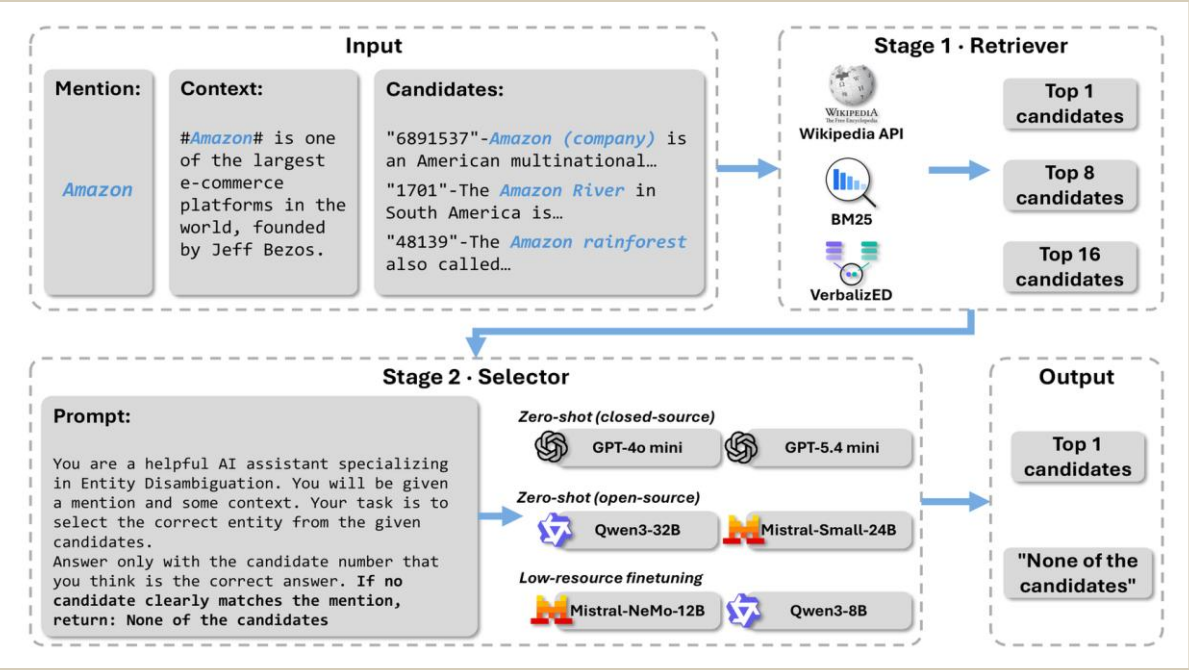


Pengyu Zhang, Yangqin Jiang, Klim Zaporojets, Congfeng Cao, Paul Groth. TimeRoute: Time-Aware Modality Routing and Diffusion for Multi-Modal Recommendation. arXiv 2608.10983, under review.

PROBLEM	Which modality matters drifts over time and differs from user to user. One global weighting cannot follow that.
METHOD	A router with fewer than 700 parameters turns each user's timestamp profile into modality weights. A time-conditioned diffusion module filters outdated edges before graph propagation.
RESULT	+5-10% over the strongest baseline on TikTok and two Amazon datasets, ten seeds. With random noise instead of timestamps the gain is gone (Amazon-Baby).

Select, Don’t Train

ISWC 2026

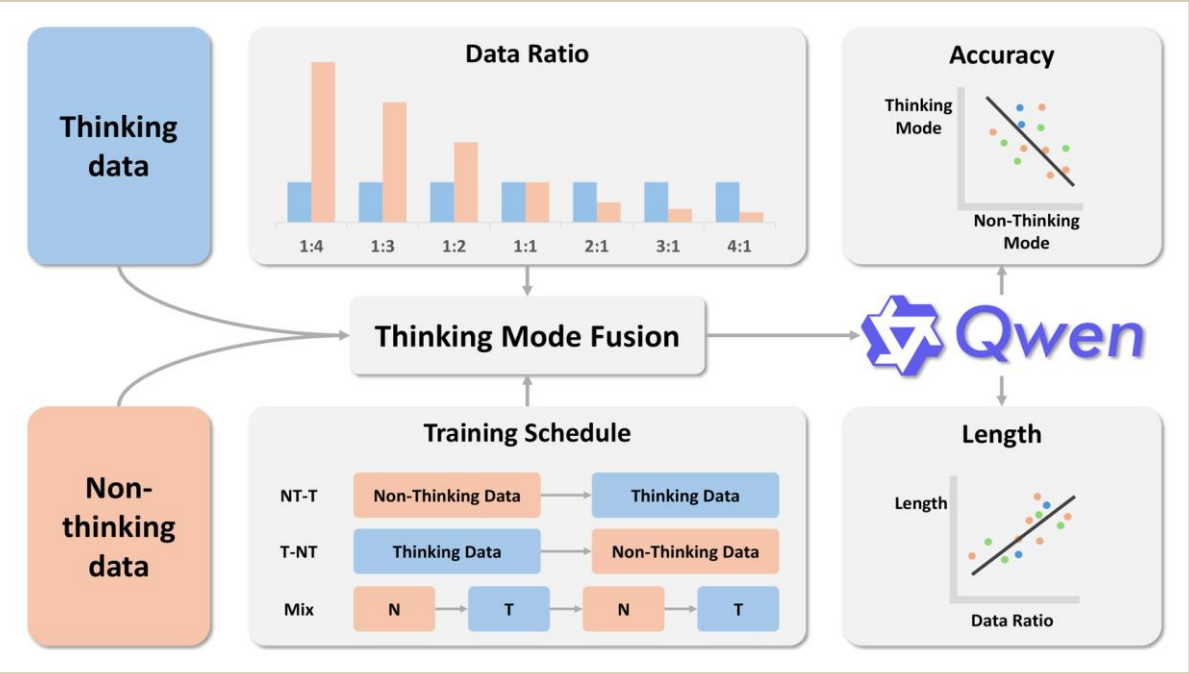


Fina Polat, Daniel Daza, Pengyu Zhang, Klim Zaporojets, Paul Groth. Select, Don’t Train: The Benefits of Modular Entity Disambiguation with LLM-Based Selection. ISWC 2026.

PROBLEM	Dual encoders put retrieval and fine-grained selection into one embedding space, and need retraining when the knowledge base changes.
METHOD	We split the two steps. A retriever returns 16 candidates, and an LLM picks one or answers that none fits.
RESULT	BM25 with GPT-5.4-mini reaches 86.3 micro-F1 on ZELDA, against 82.3 for the best dual encoder. A trained retriever raises it to 88.5. When correct abstentions count, the BM25 setup reaches 90.7.
MY PART	Third author. Fina Polat led the work, and I took part in the discussions throughout.

Fusion Training

ACL 2026 SRW

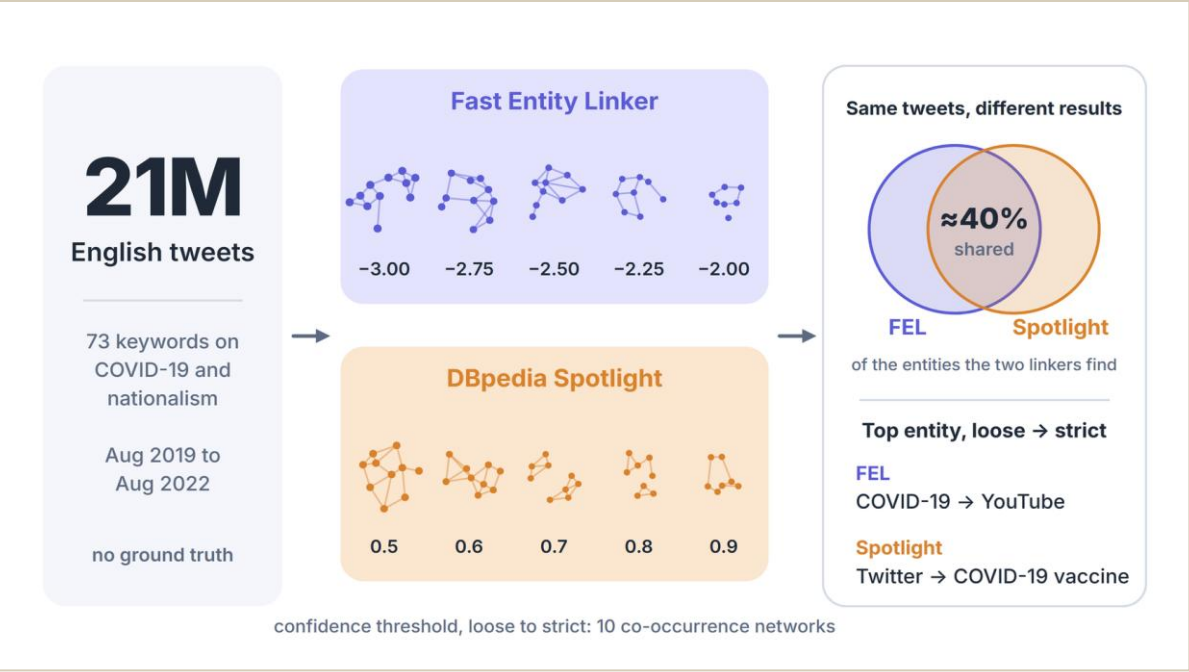


Congfeng Cao, Pengyu Zhang, Jelke Bloem. Fusion Training for Mathematical Generalization in Large Language Models. ACL 2026 Student Research Workshop.

PROBLEM	Hybrid models such as Qwen3 learn long reasoning and short answers with the same weights. How to mix the two kinds of data had not been studied systematically.
METHOD	Fusion Bench: 1,500 long-reasoning examples mixed with 375 to 6,000 short-answer ones, in seven ratios and three schedules. Qwen3-4B with LoRA, three runs each.
RESULT	More short answers lower reasoning accuracy, from 0.224 to 0.199 on average (r = -0.58). Interleaving is the most robust schedule.
MY PART	Second author. Congfeng Cao led. I helped with the logic of the paper, the figures and tables, and the revisions.

Understanding EL

EKAW 2024

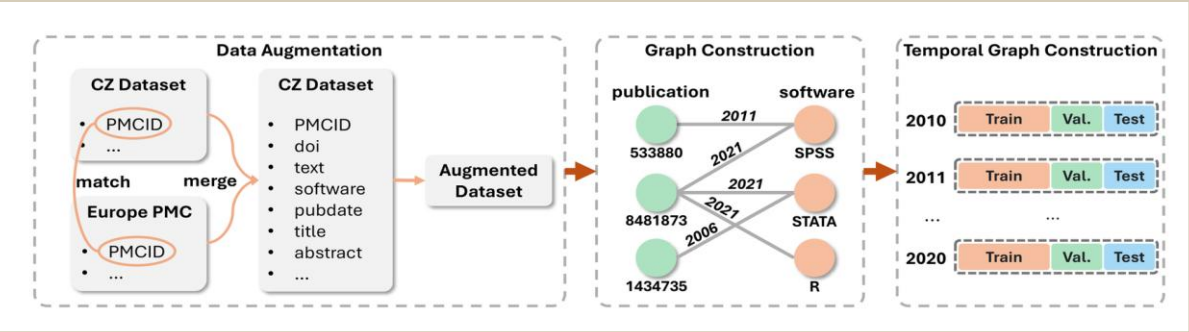


James Nevin, Pengyu Zhang, Dimitar Dimitrov, Michael Lees, Paul Groth, Stefan Dietze. Understanding the Impact of Entity Linking on the Topology of Entity Co-occurrence Networks for Social Media Analysis. EKAW 2024.

PROBLEM	For tweets there is no ground truth to choose an entity linker, but the network you build depends on the linker.
METHOD	We linked 21M tweets with FEL and DBpedia Spotlight at five thresholds each, and compared the ten co-occurrence networks.
RESULT	The two tools share about 40% of their entities (Jaccard 0.38-0.41). Size, connectivity and top entities change with tool and threshold. At strict FEL settings COVID-19 leaves the top ten.
MY PART	Second author. James Nevin led the network analysis. The entity linking pipeline was my part.

Graph-TempCZ

LREC 2026

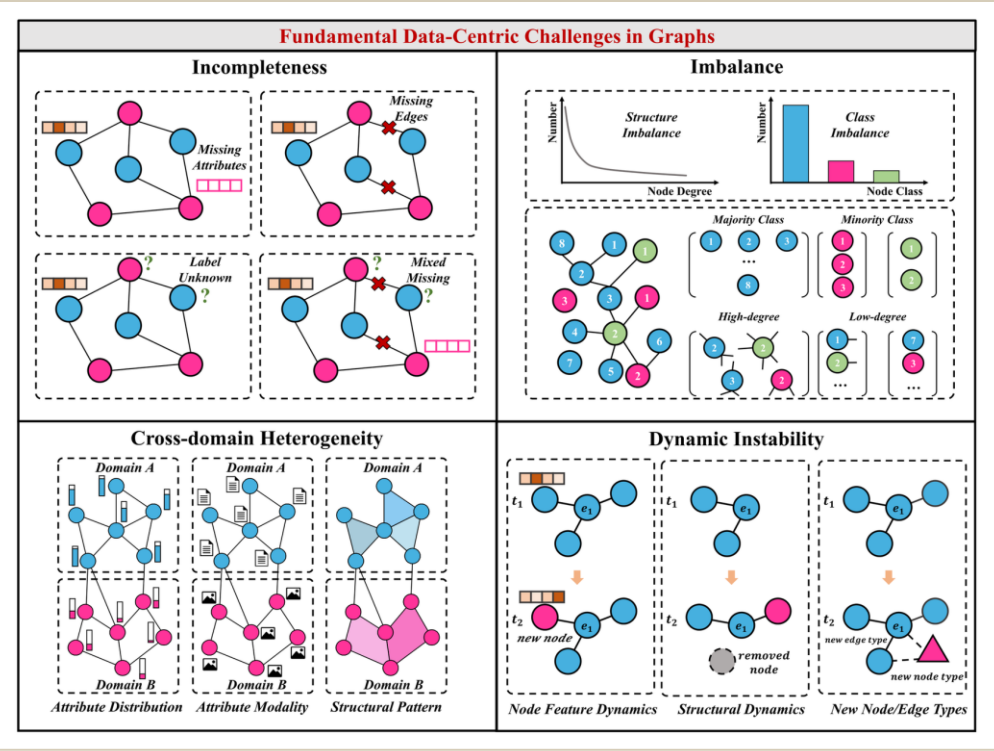


Congfeng Cao, Pengyu Zhang, Jelke Bloem. Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications. LREC 2026.

PROBLEM	Data on software use in science had no graph structure and no time axis.
METHOD	We built a publication-software graph from 6.5M software mentions in biomedical papers, and framed usage prediction as link prediction with one model per year, 2010-2020.
RESULT	GraphSAGE reaches 93% accuracy, about 6 points above the best feature-based baseline. Every model scores lower on the years after its training year.
MY PART	Second author. Congfeng Cao led. I helped with the logic of the paper, the figures and tables, and the revisions.

LLMs for graph data

ESWA 2026

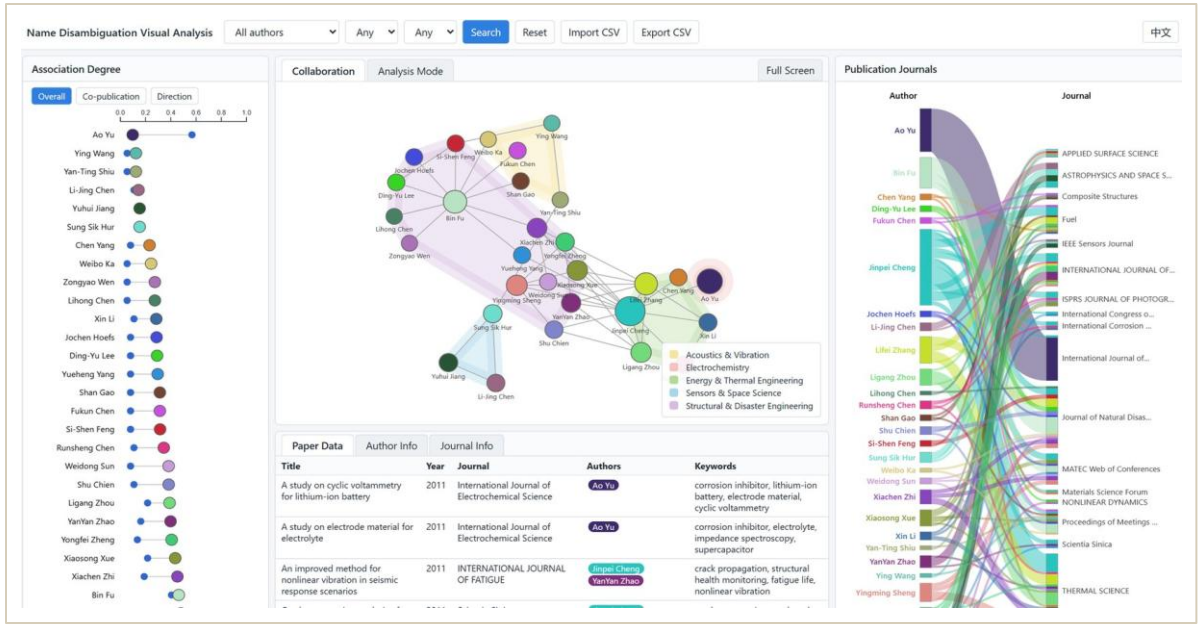


Mengran Li, Pengyu Zhang, Wenbin Xing, et al. A Survey of Large Language Models for Data Challenges in Graphs. Expert Systems with Applications, 2026.

PROBLEM	Real graph data is incomplete, imbalanced, mixed across domains and changing. Work on LLMs for graphs was not organised by these problems.
METHOD	We screened more than 1,000 papers and kept more than 380, grouped by the four data challenges, with traditional and LLM methods side by side.
RESULT	A map of the field with its open problems, and an open literature list on GitHub.
MY PART	Second author of fourteen. Mengran Li led. I wrote the parts on dynamic instability, the graphs that change over time.

Visual Analysis

JCAD 2022



Pengyu Zhang, Yong Zhang, Yanjie Cui, Baocai Yin. Visual Analysis for Name Disambiguation of Academic Papers. Journal of Computer-Aided Design & Computer Graphics, 2022 (in Chinese).

PROBLEM

Chinese names written in English collide, and existing tools were hard for research administrators to use or to check.

METHOD

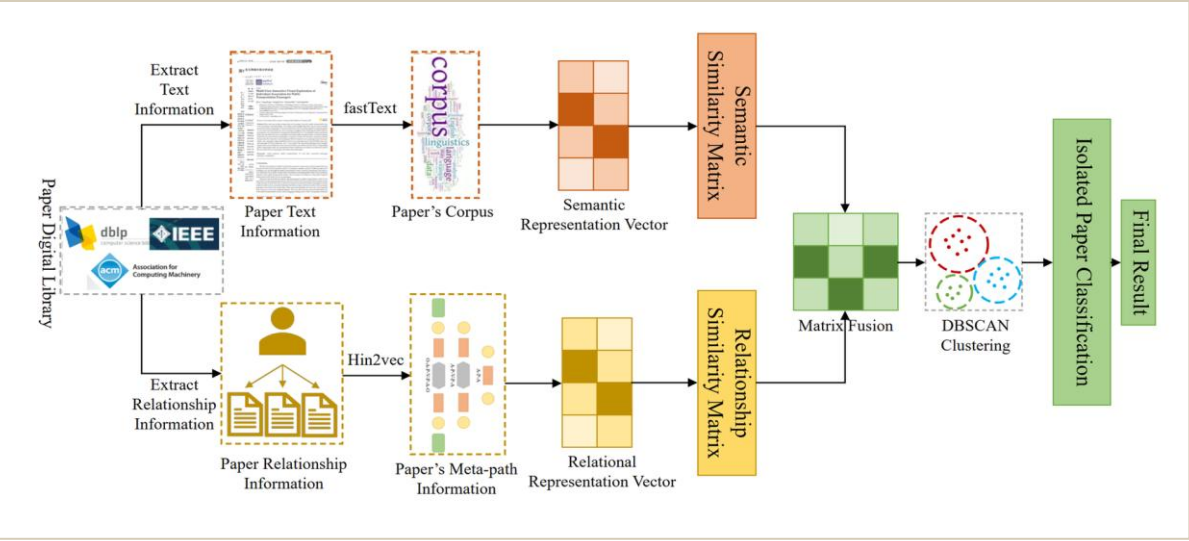
A multi-view GCN predicts each author's research field. Linked views show the collaboration network, the journals and the raw metadata, so a user can still decide when the classifier is wrong.

RESULT

The classifier reaches about 91% accuracy on ACM and DBLP. On 4,000 real university papers, eleven volunteers completed the tasks with over 95% accuracy.

DualChannel

Information 2021

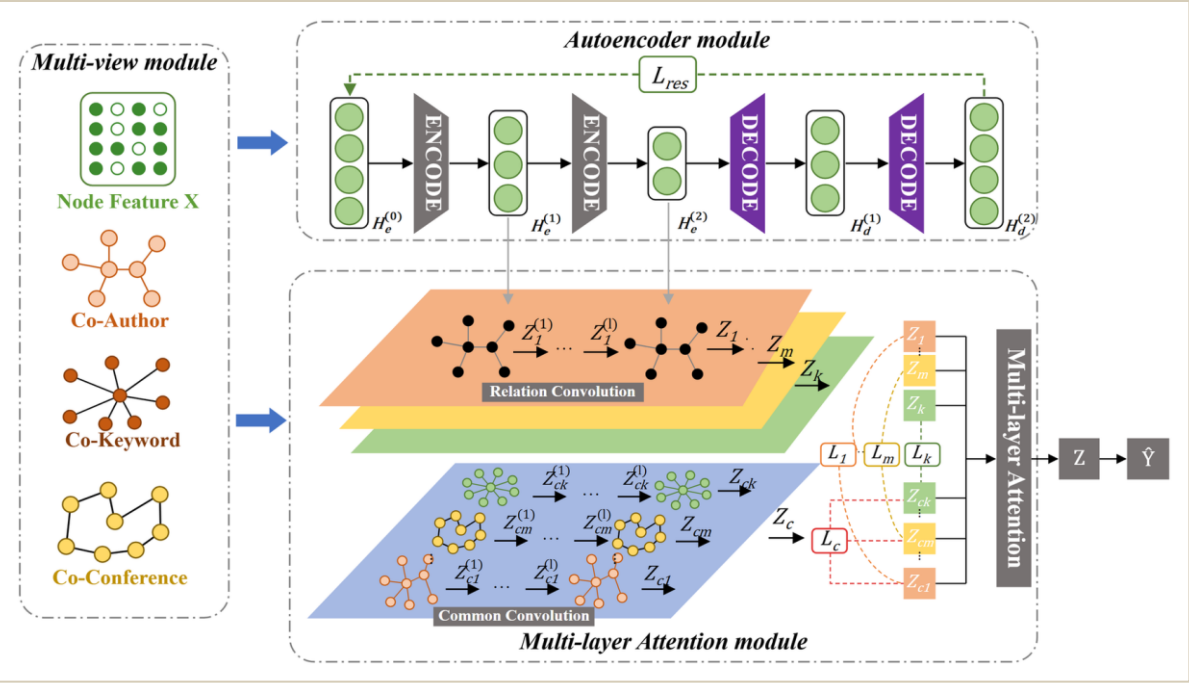


Xin Zheng, Pengyu Zhang, Yanjie Cui, Rong Du, Yong Zhang. Dual-Channel Heterogeneous Graph Network for Author Name Disambiguation. Information, 2021.

PROBLEM	Methods for author name disambiguation used either text or graph structure, rarely both.
METHOD	A text channel (fastText) and a relation channel (a co-author, organisation and venue graph with Hin2Vec). The two similarity matrices are added and clustered with DBSCAN.
RESULT	F1 0.62 on AMiner WhoIsWho, above six baselines (the best one has 0.61).
MY PART	Second author. Xin Zheng led the experiments. I wrote the manuscript and made the figures and tables.

MVMA-GCN

EAAI 2023

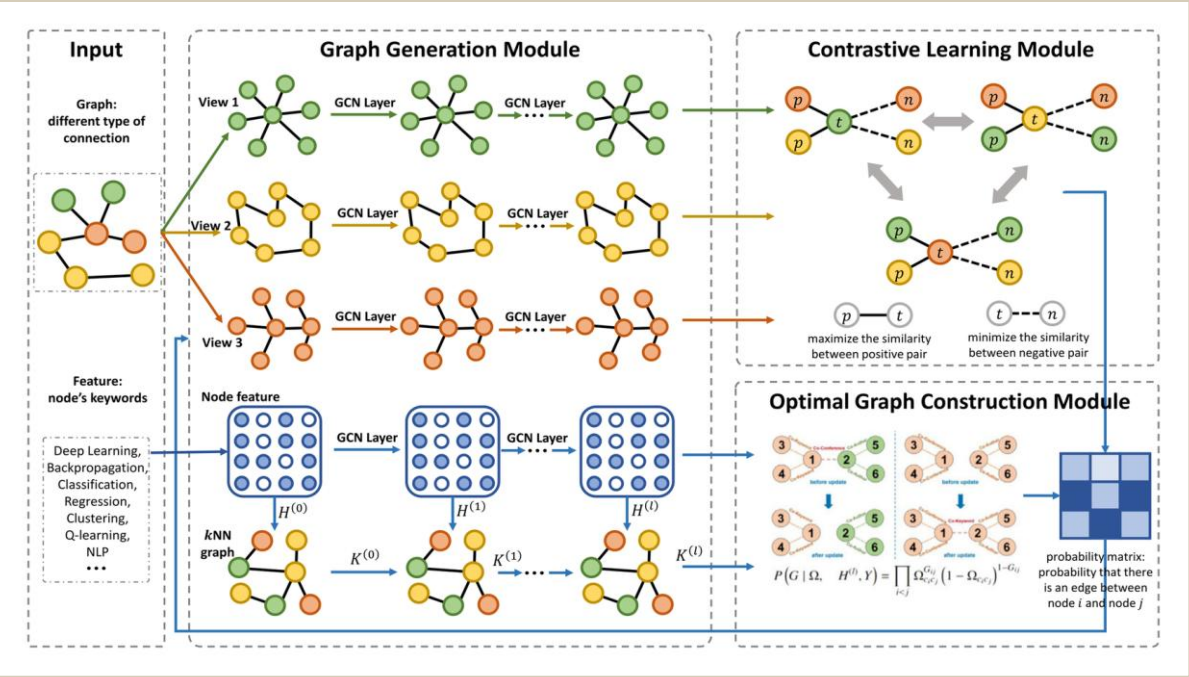


Pengyu Zhang, Yong Zhang, Jingcheng Wang, Baocai Yin. MVMA-GCN: Multi-View Multi-Layer Attention Graph Convolutional Networks. Engineering Applications of Artificial Intelligence, 2023.

PROBLEM	Real networks have several kinds of links. Single-view GNNs drop most of them and cannot weigh views or neighbours.
METHOD	All relation views and a kNN feature graph go in together. HSIC keeps the views distinct, two attention layers weigh neighbours and views, and an autoencoder keeps the features.
RESULT	Best in most settings on eight datasets. On ACM, 92.7 against 90.4 for AM-GCN with 20 labels per class.
NEXT	TIGER reuses the shared and distinct design and the HSIC loss.

RUNCL

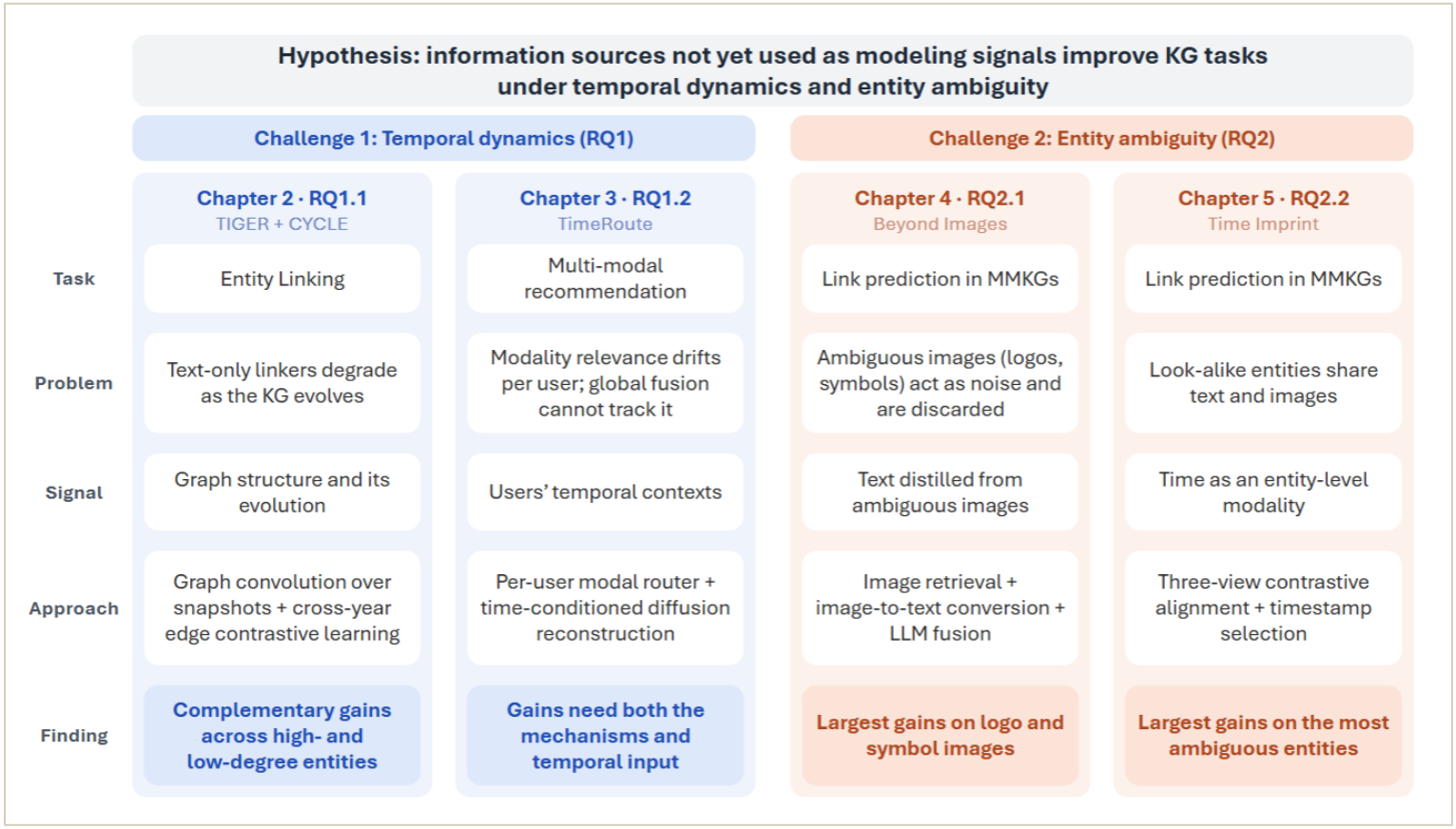
Physica A 2024



Pengyu Zhang, Yong Zhang, Xinglin Piao, Yongliang Sun, Baocai Yin. RUNCL: Relationship Updating Network with Contrastive Learning. Physica A, 2024.

PROBLEM	Missing or wrong edges hurt low-degree nodes most, but GNNs take the input graph as correct.
METHOD	Candidate kNN graphs are generated from node features, the most likely one is chosen with a stochastic block model, and the feature view and the relation view supervise each other with contrastive learning.
RESULT	Best in most few-label settings on six benchmarks. On the 200 lowest-degree test nodes of Cora, the gain over GEN grows from 3% to 6% as labels drop from 20 to 5 per class.
NEXT	CYCLE continues the focus on low-degree nodes.

Thesis overview



Pengyu Zhang. Addressing Temporal Dynamics and Entity Ambiguity in Knowledge Graph Tasks with Multi-Source Information. PhD thesis, University of Amsterdam. Defense planned for March 2027.

Open questions

1. Is one more signal worth it?
2. Are all link changes equal?
3. What breaks outside Wikidata?